

Homework 3, Sammy Abdella, sn3136

1.

1(a) No it's not valid

1(b) The argument doesn't have a valid form affirming the consequent. A counterexample would be $p = F$, $q = T$ then $p \rightarrow q$ is true and q is true but p is false

1(c) truth table for $((p \rightarrow q) \wedge q) \rightarrow p$:

p	q	$(p \rightarrow q)$	$(p \rightarrow q) \wedge q$	$((p \rightarrow q) \wedge q) \rightarrow p$
T	T	T	T	T
T	F	F	F	T
F	T	T	T	F
F	F	T	F	T

since the final column is false when $p = F$ and $q = T$ then the formula isn't a tautology which means the rule is invalid

2.

1. $rainy \rightarrow (cold \vee windy)$ Premise
2. $\neg cold$ Premise
3. $windy \rightarrow uncomfortable$ Premise
4. $\neg uncomfortable$ Premise
5. $stormy \rightarrow rainy$ Premise
6. Assume $stormy$ assumption (for contradiction)
7. $rainy$ 5, 6 Modus Ponens
8. $cold \vee windy$ 1, 7 Modus Ponens
9. $windy$ 8, 2 Disjunctive syllogism
10. $uncomfortable$ 3, 9 Modus Ponens
11. $\perp$ 10, 4 Contradiction
12. $\neg stormy$ 6–11 negation introduction (so indirect proof)

3.

3(1) $\forall x ((novel(x) \wedge bestselling(x)) \rightarrow (engaging(x) \wedge wellwritten(x)))$.

3(2) $\exists x (textbook(x) \wedge \neg difficult(x))$.

3(3) $(\forall x (biography(x) \rightarrow informative(x))) \wedge (\exists x (novel(x) \wedge bestselling(x)))$.

4.

4(a) $\forall x \exists y (comedy(y) \wedge enjoys(x, y))$.

4(b) $\exists y (drama(y) \wedge \forall x enjoys(x, y))$.

4(c) $\forall x \exists y (comedy(y) \wedge \neg enjoys(x, y)).$

4(d) $\forall x \forall y (directed(x, y) \rightarrow enjoys(x, y)).$

4(e) $\forall y (acclaimed(y) \rightarrow \exists x directed(x, y)).$

5.

5(a) $\neg(\forall x (P(x) \rightarrow Q(x))) \equiv \exists x (P(x) \wedge \neg Q(x)).$

5(b) $\neg(\exists x (R(x) \wedge \forall y S(x, y))) \equiv \forall x (\neg R(x) \vee \exists y \neg S(x, y)).$

5(c) $\neg(\forall x (\exists y P(x, y) \rightarrow \forall z Q(z))) \equiv \exists x (\exists y P(x, y) \wedge \exists z \neg Q(z)).$